GSFL: Graph-Based Spatial–Frequency Learning for Multispectral and Hyperspectral Image Fusion

Wei Chen, Maoqing Hu, Bin Yang, *Member, IEEE*, Chao Jiang, Xiaozhi Zhang, and Guofu Wang

***Abstract*—Hyperspectral and multispectral image fusion (HMIF) enhances the representational capability of remote sensing imagery by jointly exploiting spectral and spatial information. However, existing methods still struggle to jointly preserve local spatial details and global spectral consistency. Specifically, CNN-based methods capture local textures well but struggle with global spectral consistency, while Transformer-based methods leverage long-range dependencies but lack local structural inductive bias, often causing blurred fine details. In this article, we propose a graph-based spatial–frequency learning (GSFL) framework that integrates a scene-adaptive cross-modal graph (SACG) and channel-aware frequency modulation (CAFM), simultaneously enhancing spatial details and spectral consistency. Specifically, SACG introduces a cross-modal graph with graph attention, explicitly capturing structure-aware long-range dependencies for preserving spectral integrity; unlike existing graph attention network (GAT)-based methods, it adaptively determines neighborhood size using global gradients and interband Pearson correlations, enabling robust performance across diverse scenes. Meanwhile, CAFM designs a cross-domain spatial-guided frequency modulation paradigm, leveraging spatial activation intensities as content-aware priors to dynamically construct spectral gates; unlike content-agnostic static filtering or purely frequency-driven attention, it enables adaptive enhancement of critical high-frequency components while suppressing noise, effectively restoring sharp structural details. To further improve performance, we propose a complexity-aware expert transformation (CAET) built upon a differentiable mixture-of-experts (MoEs) framework to refine joint representations. In contrast to existing methods with uniform fixed transformations, CAET adaptively reweights multiple isomorphic experts based on local scene complexity, facilitating deep semantic enhancement and fine-grained feature recalibration. Comprehensive experiments on several benchmark datasets demonstrate that the proposed GSFL method achieves superior performance, surpassing several state-of-the-art methods in both quantitative metrics and qualitative evaluations, with ablation studies validating the contribution of each component. The source code and datasets are available at https://github.com/weiweiyaobaoshou/GSFL.**

***Index Terms*—Channel-aware frequency modulation (CAFM), hyperspectral and multispectral image fusion (HMIF), scene-adaptive cross-modal graph (SACG), spatial–frequency.**

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I. Introduction

HYPERSPECTRAL and multispectral image fusion (HMIF) represents a cornerstone technology in remote sensing dedicated to the reconstruction of high-resolution hyperspectral images (HR-HSIs), which are essential for diverse domains such as object detection [1], land-cover classification [2], and environmental monitoring [3], [4].

HMIF methods are primarily categorized into traditional and deep learning-based fusion methods, both of which consistently strive to enhance spatial details while preserving the spectral integrity of the fused images. Traditional fusion methods are conventionally categorized into three primary paradigms [8]: component substitution (CS), multiresolution analysis (MRA), and decomposition-based methods. CS-based methods [9] augment spatial resolution via linear transformations but often induce severe spectral distortion. MRA-based methods [10] achieve superior spectral consistency yet remain constrained by the choice of basis functions and provide suboptimal spatial detail enhancement. While matrix and tensor decomposition-based methods [11] excel in spectral preservation, they are frequently encumbered by prohibitive computational complexity and high sensitivity to noise. Overall, these traditional paradigms fail to simultaneously preserve spectral fidelity and spatial sharpness, primarily because their rigid or linear formulations cannot characterize the intricate nonlinear dependencies in heterogeneous land-cover scenes.

With the rapid development of deep learning, learning-based methods have become the mainstream due to their powerful feature representation capabilities; however, they still face challenges in simultaneously preserving structural details and spectral consistency. Fig. 1 presents representative fusion results from various paradigms, including CNN-based, Transformer-based, and spatial-frequency-based methods, as well as the proposed graph-based spatial–frequency learning (GSFL) method. CNN-based methods [5], [12] excel at extracting local textures through hierarchical convolutions, rendering them inherently proficient at recovering high-frequency spatial details. As shown in Fig. 1, CNNs produce relatively sharp edges; however, constrained by their limited local receptive fields, they struggle to capture long-range spectral dependencies [13]. This architectural limitation leads to inconsistent spectral reconstruction across global regions, as explicitly manifested by the noticeable deviation in their spectral reflectance curves.

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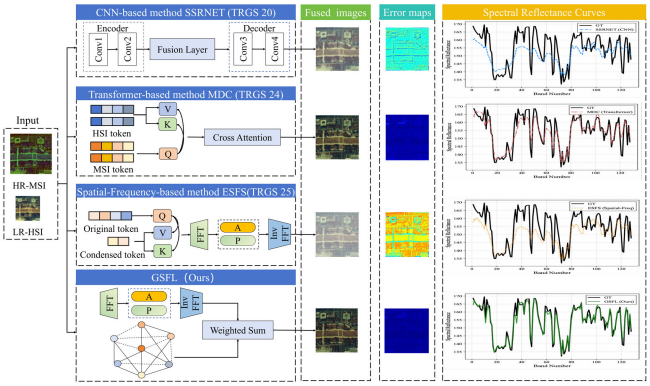


Fig. 1. Comparison of spatial detail and spectral consistency in hyperspectral image fusion across CNN-based (SSRNET [5]), Transformer-based (MDC [6]), spatial–frequency-based (ESFS [7]), and the proposed GSFL methods. Fused images and error maps visualize the recovery of spatial details, while spectral reflectance curves illustrate spectral consistency (closer alignment to the GT indicates higher fidelity).

Conversely, Transformer-based methods [6], [14] leverage attention mechanisms to capture long-range dependencies. As shown in Fig. 1, they generally outperform CNNs, with fewer macroscopic spatial residuals and better overall spectral alignment. However, they still exhibit oversmoothed local textures and minor spectral fluctuations (e.g., the deviations in MDC's spectral curve). These issues arise from the lack of local structural inductive bias [15], [16], where unguided attention propagation blurs fine details [17], and from unconstrained global token mixing, which may aggregate spectrally heterogeneous regions, limiting the ability to achieve both sharp spatial details and strict spectral fidelity.

To transcend the bottlenecks of pure spatial-domain learning, spatial-frequency dual-domain architectures have gained prominence. These methods generally fall into two categories. The first category comprises parallel-branch architectures, represented by SFINet [18], which introduce frequency priors via independent learning streams. However, constrained by position-independent static filtering, they ignore the intrinsic correlation between spatial structures and frequency components, often amplifying noise during high-frequency enhancement. The second category involves sequential-refinement paradigms, such as ESFS [7]. Although they improve feature interaction, their unidirectional staged execution (i.e., spatial augmentation followed by frequency compensation) often creates an information bottleneck, leading to the attenuation of fine-grained details before full synergy is achieved. As explicitly evidenced by the extensive spatial residuals and noticeable spectral deviations of ESFS in Fig. 1, existing dual-domain methods still struggle to effectively preserve both fine spatial details and high spectral fidelity simultaneously, necessitating a more dynamic and synergistic reconciliation paradigm.

To jointly preserve local spatial details and global spectral consistency, we propose a GSFL framework, which synergistically integrates a scene-adaptive cross-modal graph (SACG) and channel-aware frequency modulation (CAFM). Specifically, SACG constructs the cross-modal graph topology by evaluating global feature similarities to selectively connect each pixel to its top-k most relevant heterogeneous neighbors. By restricting attention to this sparse, similarity-driven neighborhood, GSFL overcomes the local receptive field constraints of CNNs to explicitly capture long-range dependencies, while avoiding the unconstrained feature blending inherent in existing Transformer architectures. Additionally, to ensure the effectiveness of the SACG across diverse terrestrial scenes, we design a scene-adaptive neighborhood mechanism. Unlike conventional fixed-topology graph methods [19], this mechanism utilizes the average global gradient and interband Pearson correlations to dynamically determine the appropriate neighborhood size, thereby preserving robust global spectral consistency across varying scene complexities.

Simultaneously, in the frequency dimension, CAFM dynamically generates band-specific enhancement coefficients by leveraging spatial-domain channel activations as explicit structural anchors. By relying on this cross-domain spatial guidance, CAFM overcomes the inherent structural blindness of existing paradigms to accurately distinguish meaningful high-frequency components from background noise. Unlike content-agnostic static filtering (e.g., SFINet [18]) that uniformly amplifies noise, or unguided frequency attention that lacks spatial constraints, this modulation mechanism ensures the selective enhancement of local spatial details, thereby achieving high-fidelity spatial reconstruction.

To further improve fusion performance, we propose a complexity-aware expert transformation (CAET) built upon a differentiable mixture-of-experts (MoEs) framework to refine joint representations. Unlike existing static paradigms [5], [20] that rely on uniform fixed transformations and struggle to accommodate local feature complexity variations, CAET utilizes a dynamic routing network to perceive local scene complexity and assigns specific response weights to a set of isomorphic experts. In addition, CAET performs weighted aggregation across all experts followed by nonlinear transformations instead of conventional sparse-selection methods [21], facilitating deep semantic enhancement and fine-grained feature recalibration. This design effectively bolsters the representation capability for complex terrestrial scenes by precisely catering to diverse feature requirements, thereby yielding high-fidelity fusion results with refined spatial structures and preserved spectral signatures.

The main contributions of this work are summarized as follows.

1) SACG constructs a scene-adaptive cross-modal graph attention network to explicitly capture long-range dependencies while effectively circumventing the unconstrained feature blending inherent in transformer-based methods, thereby maintaining strict spectral consistency.
2) CAFM establishes a cross-domain spatial-guided frequency modulation paradigm to dynamically construct spectral gates based on spatial activation intensities, while effectively overcoming the structural blindness that leads to information blurring and noise amplification, thereby achieving selective enhancement of local spatial details.
3) CAET dynamically adjusts weights based on local scene complexity instead of uniform fixed transformations to

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refine joint representations, enabling fine-grained feature recalibration and enhanced fusion quality.

The structure of this article is as follows. Section II introduces related work. Section III details the proposed GSFL method. Section IV presents the experimental setup and results. Section V concludes this article.

II. Related Work

In this section, we primarily review existing traditional methods and learning-based methods.

A. Traditional Methods

In the early stages of HMIF, the methodological framework primarily adopted pan-sharpening techniques, treating the task as an extension of multichannel structural reconstruction. Traditional methods are generally categorized into three major paradigms [8]: CS, MRA, and decomposition-based techniques. CS-based methods, including those based on IHS [22], PCA [9], [23], and Gram–Schmidt transforms, achieve rapid reconstruction by replacing spatial components with high-frequency information from multispectral images. However, the replacement mechanism may inherently lead to compromises in spectral consistency [24], [25], resulting in spectral distortion. Conversely, MRA-based methods utilize wavelets [26] or multiscale pyramids [27], [28] to infuse spatial details across multiple scales. While these methods preserve spectral integrity more effectively, they may still introduce detail bias or spatial artifacts in complex terrains due to subband response variations.

To further improve fusion quality, decomposition-based methods establish stable correlations between spectral and spatial structures through physically meaningful models. These methods primarily follow two pathways: matrix decomposition and tensor decomposition. Matrix decomposition-based methods, including CNMF [29], alternately optimize endmember and abundance matrices, while some strategies further enhance spectral retention through holographic sharpening principles [30]. Meanwhile, tensor decomposition-based methods like CSTF [31] focus on maintaining the multidimensional structural consistency of data. By incorporating nonlocal self-similarity priors [32], these methods enhance adaptability to complex textured regions. Despite these theoretical advantages, decomposition-based methods often fail to simultaneously maintain high spatial clarity and spectral precision, particularly when processing highly heterogeneous hyperspectral data.

In summary, despite their interpretability, traditional methods are limited by linear assumptions and a lack of semantic modeling, which hinders their ability to exploit complex spatial–spectral complementarity across diverse terrestrial scenarios. These constraints have driven a paradigm shift toward deep learning, where nonlinear, adaptive feature learning overcomes the representational bottlenecks of hand-crafted models to achieve a more robust synergy between spatial enhancement and spectral preservation.

B. Learning-Based Methods

1) CNN-Based Methods: With the advancement of deep learning, hyperspectral-multispectral fusion has transitioned from traditional linear models to end-to-end nonlinear mapping frameworks, with CNNs having emerged as the earliest widely adopted architectures. Early CNN-based methods typically view the fusion process as a mapping from LR-HSI and HR-MSI to HR-HSI, often operating with limited interpretability regarding their internal mechanisms. These methods feature relatively simple architectures, including the single-branch PNN [33] and 3-D-CNN [34], which simultaneously model spatial–spectral structures using 3-D convolutions. Other representative architectures, including TFNet [12] and its improved version ResTFNet, extract features from HSI and MSI, respectively, before performing cross-modal fusion. These methods advance end-to-end learning in fusion tasks, but their inherent local receptive field limitations restrict their ability to capture long-range spatial–spectral dependencies. Subsequent research introduces more complex convolutional structures and physical priors. For instance, SSRNET [5] decouples the spatial and spectral information processing using separate loss functions, enhancing the controllability of each objective. Imaging degradation models are also embedded [35] to improve interpretability. Despite the strengths of CNNs in spatial detail recovery, the inability to capture high-dimensional spectral correlations and global dependencies remains a persistent challenge.

2) Transformer-Based Methods: To overcome these limitations, Transformer-based methods have gained substantial attention for their ability to model global dependencies through attention mechanisms, effectively breaking free from local convolutional constraints. Pioneering works including Fusformer [36] and PSRT [37] demonstrate the potential of multihead attention and the hierarchical Swin Transformer for capturing long-range spatial–spectral correlations. Building on this, hybrid architectures have emerged to synergistically model local structures and global semantics. For instance, MCT [14] and UHNTC [38] integrate CNNs and Transformers into dual-branch structures to coordinate local texture with global context, while AMSFNet [39] and 3DT-Net [40], respectively, employ progressive cross-scale fusion and model-driven frameworks to further enhance edge preservation and interpretability.

Concurrently, research has focused on refining feature representation through specialized attention mechanisms and interaction strategies. DCT [41] strengthens channel relationships via directional pairwise cross-attention, while Cyformer [42] improves spectral consistency through recurrent cross-channel mechanisms. To enhance multiscale feature integration, UMSFT [43] and MDC [6] introduce deep interaction structures. Furthermore, the scope of Transformer-based fusion expands to unsupervised and generative domains [44], [45], with MSST-Net [46] utilizing self-supervised strategies and MFT-GaN [47] leveraging multiscale guidance for detail enhancement without labeled data.

Despite the competitive performance of Transformer-based architectures, they still exhibit inherent limitations in certain scenarios. Specifically, while the global attention mechanism is powerful, it often lacks the explicit structural inductive bias required for modeling the precise physical correlations between high-dimensional spectral bands [15], [16], such as manifold or graph structures. This limitation can lead to the

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loss of fine local textures or the compromise of spectral consistency in complex terrains.

3) Spatial-Frequency-Based Methods: To overcome the inherent limitations of purely spatial-domain methods in balancing spatial enhancement and spectral preservation, dual-domain frameworks have emerged to synergistically exploit both spatial and frequency domains. Initial explorations in this field, represented by SFINet [18], established a foundational parallel architecture utilizing deep Fourier transforms. However, from a technical perspective, SFINet relies on position-independent global filtering, which applies a uniform, static modulation mask across all spatial locations and spectral channels. This content-agnostic mechanism ignores the significant variations in signal-to-noise ratio (SNR) and structural richness across different hyperspectral bands, often leading to potential noise amplification in low-SNR channels or blurring in detail-intensive bands.

Subsequent research has sought to optimize parallel frameworks by incorporating advanced operators. SFIGNet [48] leverages implicit neural representation (INR) for continuous modeling; however, its implicit guidance mechanism lacks the explicit structural inductive bias required to represent complex land-cover topologies. CESFusion [49] integrates State Space Models (SSMs) to capture long-range spectral dependencies, yet its integration of frequency components follows a global-oriented interaction logic, lacking the flexibility to finely modulate dual-domain contributions according to diverse local spatial contexts.

Diverging from parallel architectures, ESFS [7] shifted toward a sequential-refinement paradigm. Although it facilitates feature reuse through residual-guided refinement and dense connections, its architectural core remains constrained by the local receptive field of standard convolutional priors. This inherent locality makes it difficult to perceive complex spatial structural correlations on a global scale, which adversely affects spectral consistency in complex scenes. Furthermore, the unidirectional sequential topology employed by ESFS, which prioritizes spatial augmentation over frequency-domain compensation, often creates a significant information bottleneck. This staged execution causes fine-grained features smoothed out during the initial spatial stage to be decoupled from subsequent frequency components, ultimately resulting in irreversible information decay.

Compared with existing dual-domain paradigms, the proposed GSFL establishes a highly synergistic framework characterized by distinct architectural features. Different from sequential pipelines like ESFS that suffer from information bottlenecks, GSFL adopts a parallel dual-branch architecture; this design facilitates the simultaneous extraction and bidirectional interaction of spatial and frequency features, thereby avoiding the irreversible attenuation of fine-grained details inherent in staged execution. For spatial modeling, unlike conventional methods relying on CNNs (limited by local receptive fields) or Transformers (prone to unguided feature blending), GSFL constructs a scene-adaptive cross-modal graph via the SACG module to explicitly capture nonlocal structure-aware dependencies, ensuring more robust global spectral consistency than traditional regular-grid encoders. Furthermore, distinguished from the content-agnostic static filtering employed in SFINet, the CAFM module in the frequency branch leverages spatial activations as content-aware priors to dynamically construct band-adaptive spectral gates; this mechanism precisely adapts to variations in SNR and structural richness across channels, achieving targeted high-frequency enhancement rather than noise-prone uniform filtering. Moreover, unlike the uniform fixed transformations [5], [20] commonly used for fusion in existing methods, the CAET module dynamically aggregates joint representations via a complexity-aware MoEs strategy, ensuring robust adaptation to heterogeneous spatial layouts. Consequently, through these mechanistic innovations, GSFL establishes a more effective and adaptive paradigm for simultaneous spatial enhancement and spectral preservation.

III. Method

A. Overview

The overall architecture of the proposed GSFL is illustrated in Fig. 2. GSFL adopts a parallel dual-branch architecture to coordinate spatial–spectral information. This parallel design enables the simultaneous extraction and mutual interaction of spatial and frequency representations, thereby effectively preventing the loss of fine-grained details. The framework mainly consists of three components: the SACG module establishes nonlocal structural dependencies to preserve spectral consistency, while the CAFM module concurrently executes adaptive high-frequency detail enhancement in the frequency domain. Subsequently, the CAET module dynamically recalibrates these fused dual-domain features based on scene complexity. The deep coupling of these three modules ensures that the framework achieves spatial–spectral synergy in a highly adaptive manner.

B. Scene-Adaptive Cross-Modal Graph

To explicitly model the long-range dependencies between HSI and MSI, SACG adopts a cross-modal graph attention network (GAT) featuring a scene-adaptive neighborhood mechanism. Unlike existing methods that rely on regular Euclidean grids for feature extraction or conventional graph-based methods with a fixed neighborhood size k, this module dynamically determines the optimal node count by calculating the average global gradient and interband Pearson correlations of the input data. Such a strategy ensures that the graph topology is adaptively tailored to the spatial–spectral complexity of each specific scene, thereby facilitating more robust feature aggregation across heterogeneous land-cover distributions.

Specifically, we first extract the deep features of the input LR-HSI and HR-MSI, where h_i^H and h_j^M represent the deep features of the ith HSI pixel and the jth MSI pixel, respectively. On this basis, as shown in Fig. 2, we calculate a global cross-modal similarity matrix to quantify the feature correlation between heterogeneous pixels. This method transcends the limitations of local receptive fields, enabling spatially distant but spectrally correlated MSI pixels to participate in feature aggregation. The dot-product similarity s_{ij} between these heterogeneous pixels is formulated as

$$s_{ij} = \left(h_i^H\right)^T h_j^M \tag{1}$$

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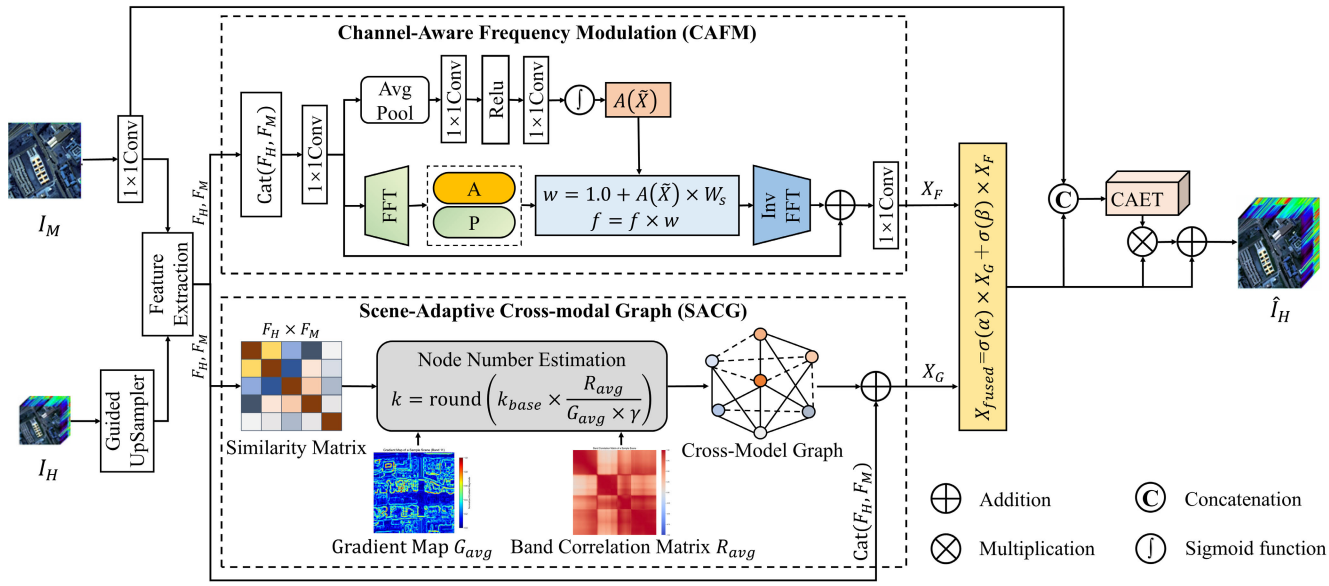


Fig. 2. Architecture of the proposed GSFL. I_M, I_H, and $\hat{I}_H$ denote the HR-MSI, LR-HSI, and the reconstructed HSI, respectively. F_M and F_H are the feature maps extracted from the HR-MSI and LR-HSI modalities. W_s represents a learnable frequency scaling parameter. CAET denotes the complexity-aware expert transformation. G_{avg} and R_{avg} are calculated from the HR-MSI and LR-HSI, respectively.

where $(\cdot)^T$ denotes the matrix transpose operation. The value of s_{ij} directly reflects the feature relevance between heterogeneous pixels, fully capturing nonlocal spatial–spectral correlations and providing a comprehensive measurement foundation for the subsequent adaptive neighbor selection.

Subsequently, we quantify the intrinsic spatial and spectral characteristics of the input scene by calculating two core dataset-level metrics. In the spatial domain, we introduce the average global gradient G_{avg} to evaluate texture richness and structural fragmentation. Given that HR-MSI provides the most precise geometric boundaries, G_{avg} is calculated on the MSI ($I_M \in \mathbb{R}^{H\times W\times B_M}$) to achieve a more accurate spatial characterization

$$G_{avg} = \frac{1}{B_M HW}\sum_{b=1}^{B_M}\sum_{x=1}^{H}\sum_{y=1}^{W}\left(\left|I_M^b(x+1,y) - I_M^b(x,y)\right| + \left|I_M^b(x,y+1) - I_M^b(x,y)\right|\right) \quad (2)$$

where $I_M^b(x,y)$ denotes the pixel value at coordinate (x,y) in the bth band. This formulation quantifies the total variation intensity by accumulating absolute finite differences across all spatial locations and spectral bands. A larger G_{avg} signifies a more fragmented land-cover distribution characterized by intricate edge information and high spatial uncertainty, reflecting lower spatial homogeneity.

Simultaneously, in the spectral domain, we calculate the mean Pearson correlation coefficient (CC) R_{avg} across all band pairs of the LR-HSI ($I_H \in \mathbb{R}^{h\times w\times B_H}$) to evaluate the interband spectral redundancy and global spectral homogeneity. Since HSI contains dozens or hundreds of contiguous bands, it provides a more holistic representation of spectral signatures compared to MSI. R_{avg} is calculated as

$$R_{avg} = \frac{2}{B_H(B_H-1)}\sum_{i=1}^{B_H}\sum_{j=i+1}^{B_H}\left|\frac{\text{cov}(b_i,b_j)}{\sigma_{b_i}\sigma_{b_j}}\right| \quad (3)$$

where b_i and b_j denote the vectorized representations of the ith and jth bands of the HSI, respectively; $\text{cov}(b_i,b_j)$ is the covariance between these two bands; and $\sigma_{b_i}, \sigma_{b_j}$ represent their respective standard deviations. R_{avg} approaching 1 indicates high spectral redundancy and strong global consistency across the scene.

To establish a rigorous mapping for k, we define two criteria: the spectral homogeneity criterion (maintaining correlation to avoid spectral distortion) and the spatial integrity criterion (modeling long-range dependencies). Mathematically, k is positively correlated with R_{avg} (high redundancy allows larger neighborhoods) and negatively correlated with G_{avg} (fragmented textures require localized neighbors to avoid feature mixing). To resolve the dimensional mismatch between the dimensionless R_{avg} and the gradient-based G_{avg}, we introduce a scaling factor γ to calibrate the denominator, formulating k as a baseline-guided ratio: $k \propto k_{base} \cdot \frac{R_{avg}}{G_{avg}\cdot\gamma}$.

As illustrated in the node number estimation mechanism in Fig. 2, the adaptive neighborhood size k is dynamically determined based on the precalculated G_{avg} and R_{avg} through a dedicated mapping function

$$k = \text{clip}\left(\text{round}\left(k_{base}\cdot\frac{R_{avg}}{G_{avg}\cdot\gamma}\right), k_{\min}, k_{\max}\right) \quad (4)$$

where k_{base} is a predefined baseline neighborhood size, and γ is a dimensionless normalization constant. Here, $k_{\min}$ and $k_{\max}$ serve as stability constraints to prevent extreme neighborhood sizes that may lead to node isolation or excessive computational overhead.

To ensure robust generalizability and prevent data leakage, γ is strictly calibrated during the training phase. For base calibration, we perform a grid search via fivefold cross-validation solely on the training set, optimizing the spectral angle mapper (SAM) metric. Once determined, γ acts as a stable, sensor-level prior. During intrascenario inference on unseen test

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regions, this precalibrated γ is directly applied, leveraging the shared statistical distributions within the same scenario. Furthermore, for cross-scenario adaptation to disparate sensors with shifted radiometric baselines, γ enables rapid domain adaptation: it can be efficiently recalibrated on a minimal target subset via the identical grid search, entirely circumventing the need to retrain the network weights.

With the similarity matrix and the estimated neighborhood size k both obtained, we formalize the cross-modal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The node set $\mathcal{V}$ is a heterogeneous set encompassing all spatial–spectral feature tokens from both the LR-HSI and HR-MSI modalities. Crucially, the edge set $\mathcal{E}$ consists of directed edges from the MSI domain to the HSI domain, as the cross-modal k-nearest neighbor (KNN) selection process is naturally asymmetric. This graph is established by selecting the top-KNNs $\mathcal{N}(i)$ globally for each HSI pixel i from MSI pixels according to s_{ij}. A multihead attention mechanism computes the attention coefficients α_{ij}

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(a^\top \left[Wh_i^H \parallel Wh_j^M\right]\right)\right)}{\sum_{m\in\mathcal{N}(i)} \exp\left(\text{LeakyReLU}\left(a^\top \left[Wh_i^H \parallel Wh_m^M\right]\right)\right)} \tag{5}$$

where W and a are learnable weight matrices and vectors, respectively, and $\parallel$ denotes the feature concatenation. To maintain spectral fidelity and training stability, node features are updated via weighted aggregation and a residual projection

$$h_i^{out} = \text{LayerNorm}\left(W_r h_i^H + \text{ReLU}\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij} W h_j^M\right)\right) \tag{6}$$

where W_r denotes the residual transformation matrix. Finally, the updated node features are rearranged into the spatial grid to form the feature map X_G.

C. Channel-Aware Frequency Modulation

To overcome the inherent structural blindness of conventional frequency-domain methods and the resulting noise amplification, the CAFM module leverages spatial-domain channel activations as explicit physical priors to dynamically construct band-specific spectral gates. Unlike rigid static filters or position-agnostic frequency attention, this cross-domain modulation mechanism selectively enhances structural details while suppressing background noise, thereby achieving targeted high-frequency detail enhancement.

Specifically, as shown in Fig. 2, the intermediate feature map $\tilde{X}$ is first projected into the frequency domain via a discrete Fourier transform (DFT). To instantiate this mechanism, a specialized spatial-domain channel attention module is employed to capture interchannel dependencies and content importance. This is achieved by applying global average pooling to the spatial features, followed by a 1×1 convolution-based bottleneck structure to generate the channel-wise modulation weights $A(\tilde{X})$. Unlike standard self-attention, $A(\tilde{X})$ serves as a cross-domain "spectral gate" derived from explicit spatial statistics, enabling the frequency-domain enhancement to be guided by underlying spatial structures. These weights

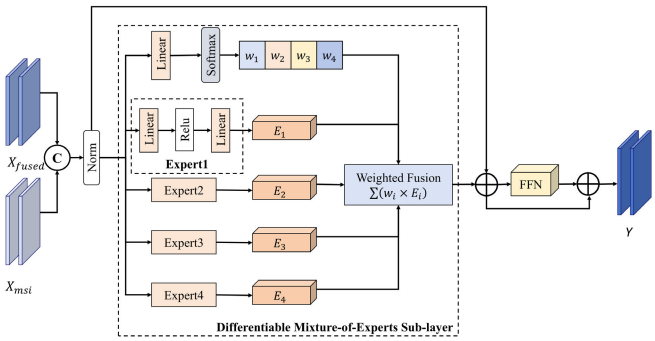


Fig. 3. Structure of the CAET.

are then integrated into the frequency space through a residual modulation scheme

$$X_F = \tilde{X} + \text{Re}\left\{\mathcal{F}^{-1}\left(\mathcal{F}\left(\tilde{X}\right) \odot \left(\mathbf{1} + A\left(\tilde{X}\right) \cdot W_s\right)\right)\right\} \tag{7}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the DFT and its inverse, respectively. $\mathbf{1}$ is an all-ones tensor and W_s represents a learnable frequency scaling parameter initialized with a truncated normal distribution $\mathcal{N}(0, 0.02^2)$ to ensure stable gradient propagation during initial training epochs. Through the residual modulation in (7), the spectral gate $A(\tilde{X})$ is injected into the frequency domain as a channel-wise scaling factor. By adopting the $\mathbf{1} + A(\cdot)$ structure, the module adaptively boosts critical structural components while preserving foundational low-frequency semantics. The operator $\text{Re}\{\cdot\}$ extracts the real component of the complex-valued inverse transform, ensuring that the modulated features X_F remain in the physical signal space. Ultimately, by utilizing spatial activation intensity as a proxy for the SNR of each band, CAFM effectively suppresses background noise in low-quality channels while precisely compensating for sharp structural details.

To exploit the robust synergy between global semantic reasoning and local detail preservation, the outputs from both modules are integrated using learnable fusion coefficients α and β

$$X_{fused} = \sigma(\alpha) \cdot X_G + \sigma(\beta) \cdot X_F \tag{8}$$

where $\sigma(\cdot)$ denotes the Sigmoid function. This adaptive mechanism allows the network to dynamically adjust the contribution of each component based on the specific requirements of regional content, ultimately generating a fused feature map with high spatial details and spectral consistency.

D. Complexity-Aware Expert Transformation

To further enhance the fusion performance, the CAET module dynamically aggregates joint spatial–spectral representations via a complexity-aware MoE strategy. Unlike rigid, uniform transformation strategies that apply a fixed mapping regardless of scene content, CAET adaptively recalibrates the fused features by accommodating the inherent variations in local structural complexity. This mechanism facilitates deep semantic refinement and ensures robust adaptation to heterogeneous spatial layouts, thereby yielding a more precise and context-aware fusion result. As illustrated in Fig. 3, CAET takes the initially fused feature maps X_{fused} and the

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channel-expanded MSI feature maps X_{msi} as joint inputs. These features are first concatenated and processed through a normalization layer to produce the stabilized joint feature $x = \text{LayerNorm}([X_{fused}, X_{msi}])$, which serves as the primary input for the subsequent transformation sublayers.

To implement adaptive transformation, the first sublayer of CAET utilizes a differentiable MoE mechanism. This component resolves spatial–spectral heterogeneity by enabling the network to dynamically shift its focus across multiple processing strategies based on the joint feature x.

Notably, the MoE paradigm in CAET is specifically tailored for pixel-level dense prediction in HMIF, distinguishing it from the discrete-routing MoE common in large-scale models. While sparse activation (e.g., Top-k) optimizes efficiency via hard assignment, it often compromises the continuous spectral consistency required for remote sensing imagery. In contrast, CAET adopts a full-expert soft aggregation mechanism, where all isomorphic experts contribute to every token via a weighted convex combination. This approach eliminates the abrupt expert switching inherent in discrete routing, realizing a smooth feature adaptation that inherently mirrors the continuous spatial–spectral variations of real-world scenes.

For an input feature token $x \in \mathbb{R}^d$, the output z_{MoE} is computed as

$$z_{MoE} = \sum_{i=1}^{N} w_i \cdot E_i(x) \tag{9}$$

where N denotes the total number of isomorphic experts, and $E_i(\cdot)$ represents the ith expert. The routing weight w_i is dynamically allocated through a Softmax function:

$$w_i = \text{Softmax}(W_r x + b_r)_i \tag{10}$$

where W_r and b_r are learnable parameters of the routing network, which map the input feature x to the expert weight space. In contrast to discrete-routing MoE, this soft aggregation mechanism provides a continuous and differentiable response to local scene complexity, effectively suppressing spatial–spectral artifacts at land-cover transitions. Furthermore, this dense activation paradigm circumvents the "routing collapse" issue typical of sparse MoE, ensuring stable gradient updates and maintaining representational diversity across experts.

Subsequently, a standard feed-forward network (FFN) is employed to map the MoE-fused features into a more abstract semantic space, complementing the specialized experts through universal nonlinear transformation. The complete CAET layer is integrated via a prenormalization residual architecture, formulated as:

$$u = x + z_{MoE}\left(\text{LayerNorm}(x)\right) \tag{11}$$

$$y = u + \text{FFN}\left(\text{LayerNorm}(u)\right). \tag{12}$$

Finally, the high-level features y are projected to generate a learned feature recalibration map Y. By element-wise multiplying this map with the preliminary fused feature maps, CAET achieves refined multisource modulation, ensuring high-fidelity detail reconstruction and superior spectral consistency.

E. Loss Function

The objective of the proposed model is to reconstruct the HR-HSI from the input LR-HSI and HR-MSI. To achieve this, the training process is supervised by two distinct loss components: a reconstruction-driven fidelity term and a feature-level regularization term.

The spectral fidelity loss $\mathcal{L}_{fidelity}$ provides the primary supervision for accurate HR-HSI reconstruction. It measures the discrepancy between the reconstructed output X_{out} and the ground-truth HR-HSI X_{ref}, encouraging the preservation of spectral signatures while maintaining spatial consistency. In this work, the Mean-Squared Error (MSE) is adopted as the fidelity loss

$$\mathcal{L}_{fidelity} = \text{MSE}\left(X_{out}, X_{ref}\right) \tag{13}$$

where X_{out} denotes the final output of the GSFL framework.

To mitigate overfitting and noise perturbations inherent in hyperspectral fusion, we introduce a diffusion feature regularization (DFR) term that enhances feature robustness by recovering clean representations from their perturbed counterparts.

Specifically, DFR is applied to each target feature map X_i, where $i = 1, 2$ denotes the MSI features (X_{msi}) and the preliminary fused features (X_{fused}), respectively. Gaussian noise $n \sim \mathcal{N}(0, \sigma^2 I)$ is added to obtain the perturbed feature $\tilde{X}_i = X_i + n$. A lightweight feature predictor $\mathcal{P}$ is then employed to reconstruct the original feature X_i from $\tilde{X}_i$. The corresponding regularization loss is defined as

$$\mathcal{L}_{DFR_i} = \text{MSE}\left(\mathcal{P}\left(\tilde{X}_i\right), X_i\right) \tag{14}$$

By minimizing the DFR loss, the network is encouraged to learn stable and noise-resilient feature representations, thereby improving generalization capability and enhancing the overall fusion quality.

Finally, the overall training objective is defined by a composite loss function $\mathcal{L}_{total}$, which integrates the aforementioned fidelity and regularization terms

$$\mathcal{L}_{total} = \mathcal{L}_{fidelity} + \lambda \cdot \left(\mathcal{L}_{DFR_1} + \mathcal{L}_{DFR_2}\right) \tag{15}$$

where λ is a hyperparameter that controls the relative contribution of the regularization terms.

IV. Experiments

A. Implementation Details

The GSFL framework is implemented using Python 3.10.13 and the PyTorch deep learning framework (with CUDA 12.1 support) on an NVIDIA GeForce RTX 4090 GPU. In the training phase, the model is optimized using the Adam optimizer with a fixed learning rate of 1×10^{-4}. The total objective function combines the mse reconstruction loss with the proposed DFR, weighted by a coefficient $\lambda = 1 \times 10^{-3}$. Additionally, for the CAET module, the number of isomorphic experts N is set to 4 across all experiments. The model is trained for a maximum of 10 000 epochs, with an early stopping mechanism based on validation performance to prevent overfitting and ensure optimal convergence. At each iteration, a random 128×128 pixel patch is cropped from the designated training area to serve as the input.

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TABLE I
QUANTITATIVE RESULTS ON THE WASHINGTON DC MALL AND URBAN DATASETS. THE BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	Washington DC Mall						Urban					
		PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	43.3511	0.7147	0.9897	1.2758	0.2224	0.4182	35.7328	0.9758	0.9945	3.2175	1.7963	3.0231
	SSRNET [5]	38.1532	0.5898	0.9665	2.3210	0.4077	0.8150	38.0035	0.9843	0.9967	2.4773	1.2745	2.4629
	HSRNET [50]	36.8523	0.5106	0.9562	2.6980	0.4670	0.7385	36.2710	0.9797	0.9952	3.0242	1.5945	2.4552
Transformer	MCTNET [14]	40.5016	0.6967	0.9805	1.7711	0.3089	0.5890	37.2357	0.9836	0.9962	2.7063	1.4484	2.5515
	MDC [6]	43.3028	0.7304	0.9897	1.2829	0.2231	0.4368	37.3509	0.9847	0.9963	2.6706	1.3910	2.4510
	AMSFNet [39]	41.1732	0.6461	0.9834	1.6393	0.2883	0.5589	33.4618	0.9636	0.9908	4.1789	2.4793	3.5363
	CYformer [42]	39.0361	0.5559	0.9726	2.0966	0.3764	0.7453	38.2956	0.9863	0.9970	2.3954	1.2605	2.4078
	CSGAV [19]	43.5624	0.6755	0.9908	1.2451	0.2217	0.4179	38.3268	0.9866	0.9970	2.3868	1.2965	2.3023
	SSDT [51]	45.8165	**0.7978**	0.9942	0.9958	0.1665	0.3139	39.2948	0.9881	0.9976	2.1420	1.0995	1.9957
Spatial-Frequency	ESFS [7]	38.3532	0.5626	0.9679	2.2682	0.4015	0.7583	37.5138	0.9848	0.9964	2.6210	1.4361	2.4045
	CaFormer [52]	41.1815	0.6662	0.9833	1.6378	0.2853	0.5600	37.0898	0.9837	0.9959	2.7521	1.4317	2.4994
	GSFL(Ours)	**47.2040**	0.7427	**0.9958**	**0.8187**	**0.1451**	**0.2952**	**39.7402**	**0.9891**	**0.9978**	**2.0284**	**1.0821**	**1.9787**

To strictly ensure evaluation integrity and prevent spatial data leakage, we follow the widely adopted experimental protocols in [5] and [6]. Specifically, the central 128 × 128 test region is permanently masked with zeros across all input modalities and ground-truth References before the training pipeline initializes and prior to applying any degradation protocols. Furthermore, the random cropping coordinates are strictly constrained to the nonmasked regions; any generated patch overlapping with the test area boundary is explicitly discarded. This rigorous spatial exclusion strategy not only ensures that unseen test pixels cannot participate in the local feature extraction of convolutional layers, the global similarity matching within the SACG, or the backward pass loss computation, effectively eliminating boundary information leakage, but also guarantees that global data statistics (e.g., min-max values for data normalization) are calculated exclusively on the valid training regions, thereby preventing any implicit statistical leakage.

In the inference phase, the trained GSFL model processes the simulated LR-HSI and HR-MSI to reconstruct the fused HR-HSI. The performance is evaluated on the central 128 × 128 test region, which remains strictly isolated and unseen during training.

B. Data and Metrics

To comprehensively evaluate the performance of our GSFL, we conduct experiments on eight publicly available datasets: Pavia Center [53], Pavia University [53], Botswana [54], Washington DC Mall [55], Urban [56], Chikusei [29], Salinas Valley Corrected [57], and Indian Pines [57]. These datasets represent diverse geographical regions and cover a wide range of spectral and spatial variations, providing a robust evaluation framework for fusion performance.

Following the general principles of Wald's protocol [58], we simulate the source images by spatially and spectrally degrading the original HR-HSI. Specifically, the spatial degradation is modeled by a 5 × 5 Gaussian filter with a standard deviation of $\sigma = 2$, followed by a 4× downsampling to generate the LR-HSI. For spectral degradation, considering the diverse sensor characteristics across datasets, we adopt a uniform spectral sampling strategy to select five representative bands from the original HR-HSI as the HR-MSI. Rather than exactly replicating specific satellite platforms, these downsampling configurations serve as a standardized benchmark for fair comparison, emulating the inherent resolution disparity of optical sensors. For practical deployment, these synthetic operators can be directly substituted with the measured MTF and SRF of target sensors to align with actual physical degradation properties. For model training and evaluation, a central 128 × 128 pixel region is designated as the test set, and the surrounding nonoverlapping areas are used for training. To ensure the integrity of the evaluation and prevent data leakage, the test region is masked with zeros prior to the training process.

In this study, six widely adopted quality metrics are used to assess the fusion performance: Peak SNR (PSNR) [59], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [60], SAM [61], CC [59], root-mean-square error (RMSE) [59], and structural similarity index (SSIM) [62]. For PSNR, CC, and SSIM, higher values indicate better fusion quality, whereas for ERGAS, SAM, and RMSE, lower values are preferred.

C. Quantitative Comparison

To validate the effectiveness of the proposed model, we conduct comparative experiments with representative methods, including CNN-based methods (TFNET [12], SSRNET [5], HSRNET [50]), Transformer-based methods (MCTNET [14], MDC [6], AMSFNet [39], CYformer [42], CSGAV [19], SSDT [51]), spatial–frequency-based methods (ESFS [7], and CaFormer [52]).

1) Washington DC Mall and Urban Datasets: Table I presents the quantitative evaluation results of various methods on the Washington DC Mall and Urban datasets. The proposed GSFL achieves the best performance in most evaluated metrics, indicating its effectiveness and robustness in diverse heterogeneous scenes. Traditional CNN-based methods (e.g., TFNET and HSRNET) show relatively limited performance, which is largely attributed to the inherent local receptive fields of convolutional kernels that may restrict the comprehensive capture of long-range spatial dependencies.

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TABLE II

QUANTITATIVE RESULTS ON THE SALINAS VALLEY AND INDIAN PINES DATASETS. THE BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	Salinas Valley						Indian Pines					
		PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	41.8589	0.8254	0.9986	1.9084	2.6440	1.4879	33.7066	0.6865	0.9937	5.2627	3.9773	3.7139
	SSRNET [5]	38.8787	0.7339	0.9995	2.6896	7.6483	2.0209	34.9951	0.6921	0.9954	4.5371	9.8915	3.3702
	HSRNET [50]	44.1116	0.8878	0.9992	1.4724	1.3790	0.9857	35.4953	0.7112	0.9958	4.2833	**2.5359**	2.8860
Transformer	MCTNET [14]	43.8823	0.8395	0.9991	1.5117	6.2471	1.4075	33.9717	0.7092	0.9941	5.1045	6.3743	3.6522
	MDC [6]	40.5405	0.8014	0.9983	2.2212	5.6663	1.6476	34.0860	0.7093	0.9942	5.0378	5.3819	3.6206
	AMSFNet [39]	42.7888	0.8361	0.9989	1.7146	3.2190	1.4851	33.5957	0.6985	0.9936	5.3303	3.8421	3.7173
	CYformer [42]	43.4341	0.7989	0.9991	1.5919	3.9189	1.3413	35.4081	0.6953	0.9957	4.4365	4.0469	3.2272
	CSGAV [19]	44.6505	0.8644	0.9993	1.3839	2.8513	1.1395	35.7757	0.7252	0.9961	4.1472	4.7607	3.1288
	SSDT [51]	45.8165	**0.9117**	**0.9996**	1.0965	**1.2402**	**0.8235**	35.3219	0.7455	0.9956	4.3605	2.6285	3.0231
Spatial-Frequency	ESFS [7]	43.1411	0.8452	0.9990	1.6465	2.6243	1.3131	35.2174	0.7572	0.9955	4.4225	3.1601	3.1858
	CaFormer [52]	42.0434	0.8317	0.9987	1.8683	4.0354	1.4038	33.0563	0.6670	0.9929	5.6719	4.3098	3.9528
	GSFL(Ours)	**46.9191**	0.8891	**0.9996**	**1.0658**	2.1263	0.9099	**36.3109**	**0.7662**	**0.9965**	**3.8994**	3.0034	**2.8391**

TABLE III

QUANTITATIVE RESULTS ON THE BOTSWANA AND CHIKUSEI DATASETS. THE BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	Botswana						Chikusei					
		PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	37.4658	0.8998	0.9979	0.4876	2.7782	2.2778	40.0885	0.9630	0.9988	0.8642	2.8607	1.6967
	SSRNET [5]	35.8141	0.7673	0.9972	0.5897	11.3419	2.8353	41.7991	0.9689	0.9995	0.5581	2.0968	1.2132
	HSRNET [50]	39.0648	0.9494	0.9986	0.4056	1.6248	1.7254	42.7842	0.7739	0.9993	0.6336	1.9837	1.2436
Transformer	MCTNET [14]	37.8378	0.8868	0.9981	0.4671	4.7848	2.1617	40.1821	0.9651	0.9980	0.8549	2.5871	1.6330
	MDC [6]	39.1259	0.9044	0.9986	0.4027	3.2376	1.9679	41.4954	0.9630	0.9988	0.8642	2.8607	1.6967
	AMSFNet [39]	37.5278	0.9064	0.9980	0.4841	3.0381	2.2593	39.7562	0.9602	0.9987	0.8979	3.0255	1.7583
	CYformer [42]	40.4697	0.9326	0.9990	0.3450	2.4306	1.7062	43.8860	0.9747	0.9995	0.5581	2.0968	1.2132
	CSGAV [19]	40.5385	0.9335	0.9990	0.3423	2.4044	1.6706	42.6197	0.9720	0.9993	0.6457	2.2831	1.3722
	SSDT [51]	39.9034	0.9383	0.9988	0.3570	2.0212	1.7249	42.0132	0.9688	0.9992	0.6885	2.4307	1.4007
Spatial-Frequency	ESFS [7]	39.3598	0.9344	0.9987	0.3920	2.6213	1.8427	41.8308	0.9693	0.9992	0.7071	2.3581	1.4410
	CaFormer [52]	38.6536	0.9145	0.9984	0.4253	2.7690	1.9251	40.5770	0.9658	0.9989	0.8169	2.5994	1.5901
	GSFL(Ours)	**41.9460**	**0.9575**	**0.9993**	**0.2911**	**1.6062**	**1.4144**	**44.4972**	**0.9764**	**0.9996**	**0.5202**	**1.9839**	**1.1457**

In contrast, Transformer-based methods exhibit substantially improved performance due to their inherent capability for global context modeling. Among them, SSDT achieves highly competitive results by utilizing a dual-branch dilated Transformer to explicitly model multiscale spatial and spectral features. However, while SSDT expands the receptive field, it inherently operates on regular Euclidean grids, which tends to constrain its flexibility in capturing intricate topological correlations in highly fragmented urban scenes. Conversely, our GSFL effectively mitigates these constraints by synergizing the SACG to model non-Euclidean long-range dependencies and the CAFM to explicitly extract high-frequency structural cues in the frequency domain. Consequently, GSFL facilitates superior spectral fidelity and sharper geometric boundaries compared to existing Transformer-based baselines.

2) Salinas Valley and Indian Pines Datasets: As shown in Table II, GSFL achieves competitive overall performance, particularly in pixel-level reconstruction (PSNR, RMSE) and structural consistency (CC). While SSDT yields favorable SSIM and ERGAS on the homogeneous Salinas dataset thanks to its fixed-rate dilated sampling, it struggles on Indian Pines. The highly fragmented landcovers and mixed pixels of Indian Pines restrict SSDT's topological adaptability, as its regular grids introduce feature interference across complex boundaries. Conversely, GSFL leverages SACG to dynamically model non-Euclidean dependencies and CAET for adaptive feature recalibration. This enables GSFL to effectively handle the extreme heterogeneity of Indian Pines while maintaining competitive spectral integrity on Salinas, achieving high-fidelity spatial–spectral preservation.

3) Botswana and Chikusei Datasets: Table III shows that GSFL achieves optimal performance across all metrics on both datasets. In Botswana's wetlands, despite employing graph-based similarity matching, CSGAV's spatial-only approach struggles with gradual spectral transitions. Conversely, GSFL accurately restores these delicate gradients via spatial-frequency coordination. On the large-scale Chikusei dataset, CYformer's cyclic iterative refinement yields competitive results, yet neglecting frequency-domain correlations limits its reconstruction of dense architectural textures. Similarly, while SSRNET's 3-D-CNN maintains local consistency, its restricted receptive field hinders high-frequency detail restoration. Overall, GSFL demonstrates exceptional adaptability to both natural spectral gradients and complex urban-agricultural spatial heterogeneities.

4) Pavia Center and Pavia University Datasets: As shown in Table IV, GSFL achieves the best overall performance. While Transformer-based methods (e.g., CYformer,

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TABLE IV

QUANTITATIVE RESULTS ON THE PAVIA CENTER AND PAVIA UNIVERSITY DATASETS. THE BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	Pavia Center						Pavia University					
		PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	35.4458	0.9762	0.9931	4.1303	4.6620	4.7535	40.7880	0.9687	0.9961	2.2459	1.6003	2.4964
	SSRNET [5]	37.4458	0.9833	0.9956	3.4217	3.9194	3.8351	43.1551	0.9803	0.9977	1.7101	1.2531	1.9986
	HSRNET [50]	36.7543	0.9809	0.9944	3.7053	4.0922	4.3072	40.9570	0.9758	0.9966	2.2026	1.5126	2.2142
Transformer	MCTNET [14]	39.0559	0.9862	0.9968	2.8428	3.2185	3.6854	42.8650	0.9787	0.9976	1.7682	1.2892	2.0786
	MDC [6]	37.9539	0.9834	0.9958	3.2274	3.6238	4.1367	42.7768	0.9781	0.9975	1.7863	1.3129	2.1047
	AMSFNet [39]	37.9764	0.9799	0.9957	3.2190	3.5760	4.3545	38.0945	0.9566	0.9932	3.0624	2.0406	2.9983
	CYformer [42]	39.3800	0.9863	0.9970	2.7387	3.1178	3.6870	44.0850	0.9821	**0.9982**	1.5365	1.1556	1.8376
	CSGAV [19]	39.7024	0.9865	0.9972	2.6389	3.0443	3.6654	43.3702	0.9811	0.9979	1.6683	1.2173	1.9662
	SSDT [51]	39.6883	0.9866	0.9971	2.6520	2.9788	3.6396	43.9771	0.9821	0.9981	1.5555	1.1644	1.8491
Spatial-Frequency	ESFS [7]	37.2944	0.9831	0.9953	3.4819	3.9328	4.1250	42.2195	0.9794	0.9973	1.9046	1.3673	2.1017
	CaFormer [52]	37.0000	0.9814	0.9949	3.6020	4.0368	4.0791	41.6124	0.9771	0.9969	2.0425	1.4261	2.1084
	GSFL(Ours)	**39.9152**	**0.9871**	**0.9973**	**2.5750**	**2.8767**	**3.5986**	**44.3240**	**0.9824**	**0.9982**	**1.4948**	**1.1292**	**1.8153**

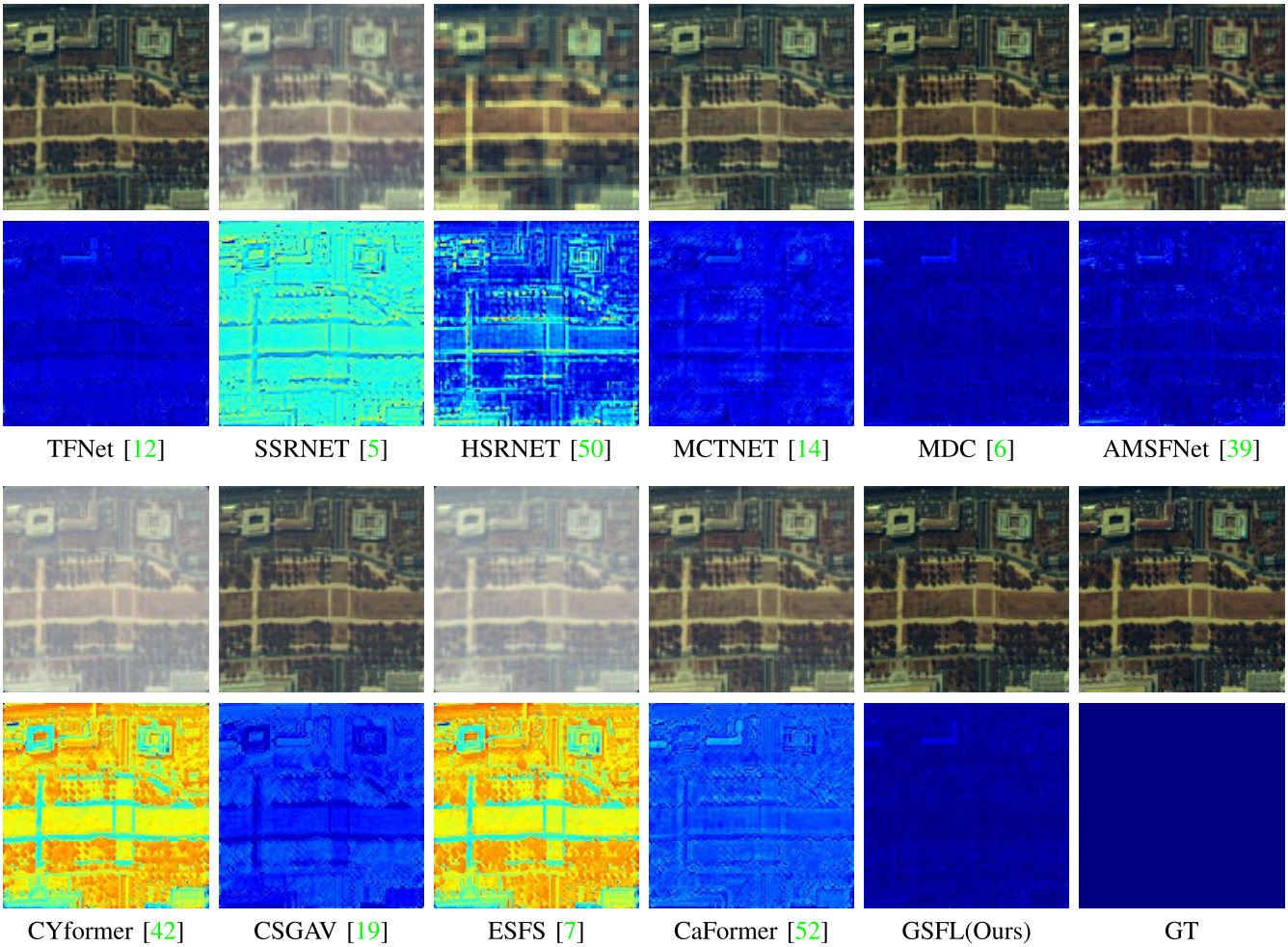


Fig. 4. Fusion results of different fusion methods on the Washington DC Mall dataset. The first row displays the pseudo-color RGB images of the fusion results, while the second row presents the corresponding absolute error maps.

SSDT) excel at geometric modeling, their uniform processing often faces bottlenecks at the abrupt local transitions (e.g., roof-to-shadow boundaries) typical of dense urban Pavia scenes. In contrast, GSFL leverages the CAET to adaptively differentiate processing strategies—preserving spectral integrity in homogeneous pavements while effectively enhancing high-frequency details at intricate boundaries. This sample-adaptive recalibration enables GSFL to resolve dense urban textures while effectively preserving spectral signatures.

D. Qualitative Comparison

1) Washington DC Mall and Indian Pines Datasets: Figs. 4 and 5 present the pseudo-color RGB images and absolute error maps for the Washington DC Mall and Indian Pines datasets,

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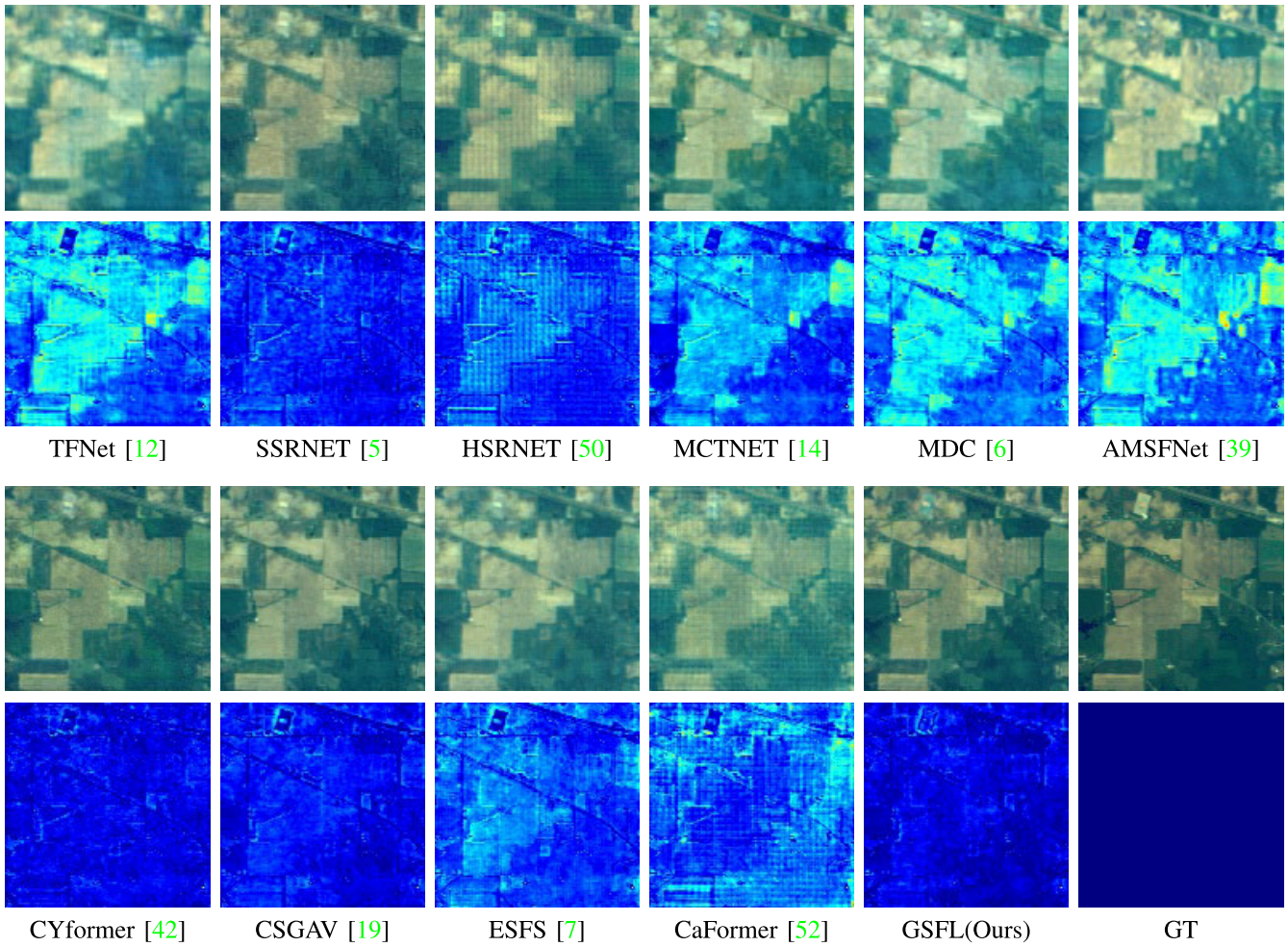


Fig. 5. Fusion results of different fusion methods on the Indian Pines dataset. The first row displays the pseudo-color RGB images of the fusion results, while the second row presents the corresponding absolute error maps.

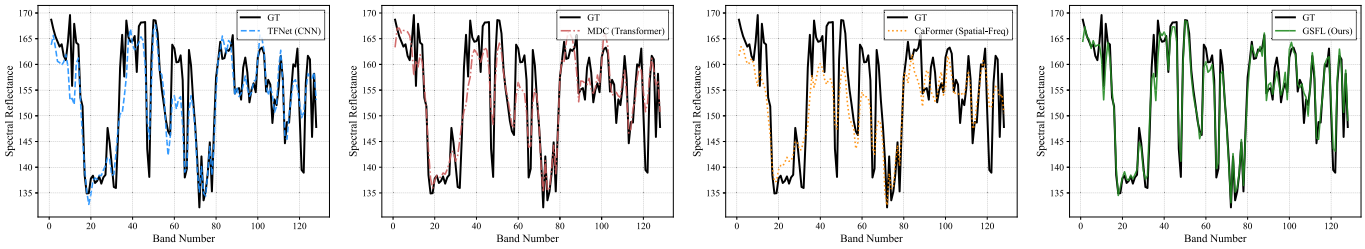


Fig. 6. Comparison of spectral reflectance curves at coordinate (62, 37) on the Washington DC Mall dataset. The black solid line denotes the GT. (a) TFNet [12]. (b) MDC [6]. (c) CaFormer [52]. Our GSFL (d) achieves the highest fidelity to the original spectral signatures.

representing urban and agricultural landscapes, respectively. In the complex urban environment of Washington DC, methods like CSGAV and HSRNET exhibit yellow residuals along building edges, indicating a lack of specialized high-frequency processing, while SSRNET and CYformer suffer from global spectral degradation and "gray veil" artifacts. Similarly, in the highly fragmented Indian Pines scene, models such as TFNet and AMSFNet fail to maintain spectral consistency across diverse agricultural plots. While Transformer-based models (e.g., CSGAV, CYformer) improve structural coherence, they still leave residual outlines in dense vegetation transition zones. In contrast, the proposed GSFL achieves excellent visual clarity and accurate reconstruction across both datasets, yielding remarkably clean, nearly uniform dark-blue error maps. Notably, it effectively preserves fine rooftop textures in urban areas and interclass discriminative details in severely mixed agricultural regions. This verifies that GSFL's bidirectional spatial-frequency coordination adaptively enhances high-frequency structural signals while minimizing spectral distortions, ensuring exceptional geometric and spectral fidelity in highly heterogeneous environments.

2) Spectral Consistency: To evaluate microscopic fidelity, Fig. 6 compares the spectral reflectance curves of GSFL against representatives from CNN (TFNet), Transformer (MDC), and hybrid (CaFormer) architectures at coordinate (62, 37). While competing methods exhibit noticeable spectral

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TABLE V
METRICS OF CLASSIFICATION RESULTS ON PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Method	OA(%)↑	AA(%)↑	Kappa↑	mIoU↑	F1↑
Baseline	A2S2K-ResNet [63]	98.94±0.8486	98.83±0.9141	0.9860±0.0112	0.9779±0.0151	0.9882±0.0082
CNN	TFNet [12]	99.78±0.0379	99.77±0.0261	0.9971±0.0005	0.9937±0.0011	0.9968±0.0005
	SSRNET [5]	99.63±0.1694	99.51±0.2523	0.9952±0.0022	0.9899±0.0044	0.9949±0.0022
	HSRNET [50]	99.78±0.0015	99.78±0.0264	0.9970±0.0001	0.9939±0.0004	0.9969±0.0002
Transformer	MCTNET [14]	98.94±0.7756	98.36±1.2971	0.9860±0.0103	0.9729±0.0187	0.9856±0.0101
	MDC [6]	99.80±0.0262	99.79±0.0416	0.9974±0.0003	0.9940±0.0015	0.9970±0.0007
	AMSFNet [39]	99.72±0.0321	99.58±0.0913	0.9963±0.0004	0.9894±0.0011	0.9946±0.0005
	CYformer [42]	99.77±0.0365	99.79±0.0527	0.9970±0.0005	0.9942±0.0011	0.9971±0.0005
	CSGAV [19]	99.79±0.0044	99.75±0.0013	0.9972±0.0001	0.9937±0.0002	0.9968±0.0001
Spatial-Frequency	ESFS [7]	99.64±0.1197	99.60±0.1388	0.9953±0.0016	0.9892±0.0036	0.9945±0.0018
	CaFormer [52]	99.40±0.3622	99.17±0.5626	0.9921±0.0048	0.9846±0.0088	0.9921±0.0046
	GSFL(Ours)	**99.81±0.0146**	**99.80±0.0337**	**0.9974±0.0002**	**0.9945±0.0010**	**0.9972±0.0005**

bias and oversmoothing in sharp mutation regions (e.g., bands 50–70), GSFL achieves the highest congruence with the ground truth (GT). This validates that our CAFM and SACG modules effectively preserve fine-grained spectral profiles without introducing artifacts.

E. Classification Performance Evaluation

To verify the effectiveness of the proposed method in downstream applications, hyperspectral image classification experiments are conducted on the Pavia University dataset. The state-of-the-art remote sensing classification model A2S2K-ResNet [63] is employed as the baseline classifier. For the experimental configuration, 90% of the samples are randomly allocated for training and validation, while the remaining 10% are used for testing. The fusion results of the proposed GSFL are compared with several representative methods across three categories: CNN-based methods (TFNet, SSRNET, and HSRNET), Transformer-based methods (MCTNET, MDC, AMSFNet, CYformer, and CSGAV), and spatial-frequency-based methods (ESFS and CaFormer). The classification performance is quantitatively assessed using five metrics: overall accuracy (OA), average accuracy (AA), the Kappa coefficient (Kappa), mean intersection over union (mIoU), and the $F1$-score [64], [65], [66].

Table V summarizes the quantitative classification results. As shown, the proposed GSFL method achieves the highest scores across all evaluation metrics. Compared to the baseline A2S2K-ResNet, all fusion-based methods exhibit noticeable improvements, demonstrating the necessity of data fusion for enhancing downstream task performance. Among the competitors, Transformer-based models like MDC and CYformer achieve competitive but inferior performance, benefiting from their capacity to model long-range dependencies. In contrast, GSFL generally outperforms these methods, particularly in the mIoU and $F1$-score metrics, which reflect better class-specific discriminative power and regional consistency. Furthermore, compared to other spatial-frequency methods including ESFS and CaFormer, GSFL shows a notable performance gain.

TABLE VI
ABLATION STUDY RESULTS ON WASHINGTON DC MALL DATASET, WHERE "W/O" DENOTES TRAINING WITHOUT THE SPECIFIED COMPONENT. BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Module	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
W/O SACG	45.271	0.728	0.994	1.023	0.181	0.346
W/O CAFM	44.446	0.700	0.992	1.125	0.198	0.378
CAFM W/O Attn	46.251	0.719	0.994	0.913	0.161	0.323
CNN+CAFM	45.111	0.700	0.994	1.042	0.184	0.357
Transformer+CAFM	45.704	0.736	0.994	0.923	0.171	0.333
W/O CAET	44.356	0.698	0.992	1.136	0.201	0.397
GSFL-SACG(Fixed k)	45.622	0.709	0.994	0.982	0.173	0.337
GSFL	**47.204**	**0.739**	**0.996**	**0.819**	**0.145**	**0.295**

This effectiveness stems from the dual-domain coordination of GSFL, which ensures that both sharp geometric structures and authentic spectral signatures are preserved during fusion, thereby providing more reliable input for the classifier. These results demonstrate the potential of the proposed GSFL in supporting high-precision remote sensing downstream tasks.

F. Ablation Study

To validate the effectiveness of each component in the proposed GSFL framework, comprehensive ablation studies are conducted on the Washington DC Mall dataset. Each row in Table VI represents a specific structural modification: "W/O SACG" indicates the removal of the SACG, leaving only the frequency-domain modulation and convolutional features; "W/O CAFM" indicates the removal of the CAFM; "CAFM W/O Attn" represents a fine-grained degraded variant where the core attention mechanism within the CAFM is removed (leaving only basic convolutional or linear projection operations); "CNN+CAFM" and "Transformer+CAFM" represent replacing the SACG with standard 3-D convolutions or a standard self-attention module, respectively; For the "Transformer+CAFM" variant, its core configuration is strictly aligned with the full GSFL framework to ensure a fair comparison: it adopts an eight-head multihead self-attention module

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TABLE VII
QUANTITATIVE EVALUATION ON PAVIA UNIVERSITY DATASET AT SCALE FACTOR 8. BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	39.2648	0.6058	0.9740	2.0280	0.3573	0.6774
	SSRNET [5]	37.4645	0.5929	0.9599	2.5125	0.4401	0.8983
	HSRNET [50]	37.8092	0.5667	0.9633	2.4147	0.4187	0.7936
Transformer	MCTNET [14]	42.5430	**0.7107**	0.9881	1.4002	0.2435	0.4569
	MDC [6]	42.4241	0.7023	0.9869	1.4312	0.2510	0.4689
	AMSFNet [39]	39.4266	0.5874	0.9759	2.0079	0.3514	0.6644
	CYformer [42]	39.5713	0.5572	0.9753	1.9714	0.3537	0.6940
	CSGAV [19]	41.9124	0.6374	0.9862	1.5056	0.2664	0.5329
	SSDT [51]	42.7218	0.6887	0.9882	1.3770	0.2395	0.4576
Spatial-Frequency	ESFS [7]	41.7032	0.6580	0.9864	1.6032	0.2741	0.5891
	CaFormer [52]	42.1155	0.6951	0.9871	1.4708	0.2562	0.4839
	GSFL (Ours)	**43.1379**	0.6769	**0.9893**	**1.3075**	**0.2293**	**0.4488**

with 64-D per-head features, an 8×8 local window, a two-layer FFN with GELU activation and 256 hidden dimensions, and follows the same prenormalization residual architecture as the full model. "W/O CAET" indicates the removal of the CAET; "GSFL-SACG (Fixed k)" denotes the variant that retains the entire cross-modal graph structure of SACG but uses a fixed neighborhood size ($k = 5$) instead of the proposed scene-adaptive mechanism. The selection of $k = 5$ follows the conventional configuration established in existing graph-based methods, such as CSGAV [19]. The performance of the GSFL serves as the benchmark.

As shown in Table VI, removing the graph module ("W/O SACG") causes a noticeable PSNR drop (1.93 dB), demonstrating that explicitly modeling long-range non-Euclidean dependencies is crucial for capturing global spatial–spectral contexts. Replacing SACG with CNNs or Transformers ("CNN+CAFM" and "Transformer+CAFM") yields higher SAM/ERGAS errors; this confirms that, unlike unguided standard self-attention or localized convolutions, our graph-based structural guidance effectively mitigates spectral blurring around material boundaries. Furthermore, "W/O CAFM" degrades performance by removing adaptive frequency-domain modulation, which is vital for recovering high-frequency structural details. Similarly, "CAFM W/O Attn" confirms that content-adaptive frequency modulation is consistently superior to static spectral filtering, indicating that dynamic weights are essential for accurately preserving structural consistency and overall signal fidelity. The severe degradation in "W/O CAET" highlights the indispensable role of this dynamic nonlinear module in fine-tuning feature representations for joint spatial–spectral enhancement. Finally, the inferior performance of "GSFL-SACG (Fixed k)" explicitly verifies that our data-driven adaptive neighborhood mechanism effectively aligns graph topologies with real scene complexities. Ultimately, all ablation variants consistently underperform the full GSFL, validating the synergistic necessity of our proposed components.

G. Robustness and Generalization Analysis

To demonstrate the practical applicability of the proposed GSFL, we comprehensively evaluate its generalization capabilities across varying scales and dimensions, as well as its robustness against complex data degradations.

TABLE VIII
QUANTITATIVE EVALUATION OF THE PROPOSED GSFL AND COMPARISON METHODS ACROSS DIFFERENT SPATIAL DIMENSIONS ON WASHINGTON DC MALL DATASET. BEST RESULTS ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Category	Methods	PSNR↑	SSIM↑	CC↑	RMSE↓	ERGAS↓	SAM↓
CNN	TFNET [12]	42.5626	0.7059	0.9899	1.2828	0.2210	0.4193
Transformer	MCTNET [14]	43.4288	0.7552	0.9928	1.1611	0.1988	0.3836
	CSGAV [19]	42.3826	0.6797	0.9903	1.3097	0.2304	0.4243
	SSDT [51]	44.8076	**0.8014**	0.9941	0.9945	0.1696	0.3186
Spatial-Frequency	CaFormer [52]	40.3084	0.6616	0.9838	1.6629	0.2863	0.5711
	GSFL (Ours)	**45.9760**	0.7241	**0.9955**	**0.8659**	**0.1514**	**0.3002**

1) Cross-Scale Generalization: To evaluate model adaptability under varying sensor resolutions, we conduct a cross-scale generalization test on the Pavia University dataset at a challenging $8\times$ scale factor. As shown in Table VII, the extreme downsampling factor causes substantial performance degradation across all methods. However, GSFL demonstrates superior resilience, securing optimal values in most metrics despite severe resolution loss. Notably, while some methods (e.g., MCTNET and SSDT) yield slightly higher SSIM, their elevated SAM and ERGAS values indicate that their structural enhancements coincide with unintended spectral deviations. In contrast, GSFL better captures the intrinsic spatial-frequency characteristics that remain stable against severe resolution loss. Its dual-path coordination preserves superior spatial–spectral integrity, indicating the learning of robust and highly generalizable fusion priors.

2) Spatial Dimension Generalization: To simulate variable observation swaths in real-world scenarios, we evaluate our model (trained on 128×128 patches) directly on 64×64 regions. As shown in Table VIII, GSFL exhibits superior stability compared to the baseline models across different spatial scales. Furthermore, to resolve tensor mismatches in nonstandard large-scale images (e.g., 503×503), we implement a standardized padding-and-cropping inference pipeline. The input is symmetrically padded to satisfy the intrinsic downsampling factor, inferred via overlap-tiling with Hann

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TABLE IX

PSNR, ERGAS, SAM, AND SSIM RESULTS OF DIFFERENT FUSION METHODS UNDER DIFFERENT GAUSSIAN NOISE LEVELS ON THE PAVIAU DATASET (THE BEST VALUES ARE IN BOLD AND THE SUBOPTIMAL RESULTS ARE UNDERLINED)

Category	Methods	20dB				30dB				40dB			
		PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
CNN	TFNET [12]	34.0610	0.8890	3.0648	6.2375	40.0796	0.9673	1.6843	2.6107	40.0856	0.9673	1.6827	2.6071
	SSRNET [5]	39.2687	0.9627	1.7483	3.1361	41.7210	0.9769	1.3947	2.2753	42.0693	0.9784	1.3531	2.1623
	HSRNET [50]	32.6364	0.8466	3.6020	7.1361	38.5393	0.9577	1.9032	3.2430	40.0433	0.9723	1.6390	2.4505
Transformer	MCTNET [14]	41.0069	**0.9770**	1.4959	**2.3494**	41.0594	0.9772	1.4861	2.3286	41.0535	0.9772	1.4854	2.3284
	MDC [6]	40.9667	0.9735	1.4912	2.4048	41.5351	0.9763	1.4215	2.2619	41.5822	0.9765	1.4167	2.2519
	AMSFNet [39]	37.5630	0.9555	2.1024	3.0639	37.5632	0.9555	2.1009	3.0617	37.5566	0.9554	2.1021	3.0632
	CYformer [42]	39.4290	0.9608	1.7233	3.2781	41.0467	0.9695	1.4992	2.7099	42.5774	0.9795	1.2750	2.0538
	CSGAV [19]	38.1272	0.9504	1.9715	3.8127	41.6588	0.9762	1.3923	2.3729	42.2371	0.9789	1.3251	2.1799
	SSDT [51]	40.0353	0.9671	1.6912	2.6286	42.3570	0.9783	1.2992	2.1816	42.7971	**0.9801**	1.2491	2.0304
Spatial-Frequency	ESFS [7]	39.9313	0.9686	1.6677	2.8049	41.1733	0.9767	1.4863	2.3105	41.3236	0.9775	1.4655	2.2506
	CaFormer [52]	38.8157	0.9616	1.8164	2.8049	40.4149	0.9736	1.5567	2.3130	40.6246	0.9749	1.5245	2.2555
	GSFL(Ours)	**41.4868**	0.9746	**1.4042**	2.3830	**42.4771**	**0.9790**	**1.2861**	**2.0834**	**43.3076**	0.9794	**1.2327**	**1.9996**

window blending to suppress boundary artifacts, and deterministically cropped to restore original dimensions. This strategy ensures the reliable processing of diverse large-scale data in practical deployments.

3) Degradation Robustness: To evaluate practical reliability under inevitable imaging noise, we conduct comparative experiments on the PaviaU dataset (4× scale) across varying Gaussian noise levels (SNR = 20, 30, and 40 dB). As summarized in Table IX, although noisy conditions naturally degrade fusion quality, GSFL consistently maintains the highest stability, outperforming top-tier competitors like MCTNET and SSDT by a clear margin, particularly under extreme 20 dB noise. This superior noise immunity stems from its dual-domain modeling: the frequency modulation in CAFM effectively suppresses random noise, while the graph-based aggregation in SACG prevents local artifacts from spreading globally. These results confirm that GSFL possesses robust noise resistance and reliable generalization for nonideal satellite scenarios.

V. CONCLUSION

In this article, we propose a spatial-frequency domain fusion framework named GSFL to jointly improve spatial detail reconstruction and spectral fidelity. The SACG establishes structure-aware long-range dependencies via a scene-adaptive graph mechanism in non-Euclidean space, effectively alleviating the inherent limitations of constrained receptive fields and unconstrained information blending in conventional architectures. Concurrently, the CAFM overcomes the structural blindness inherent in conventional static filtering and purely frequency-driven attention. By dynamically modulating critical frequency components via spatial guidance, it effectively mitigates information blurring and noise amplification, achieving fine spatial structure restoration. Furthermore, the CAET performs fine-grained feature recalibration according to local scene complexity, ensuring deep semantic enhancement across diverse terrestrial landscapes. Extensive evaluations demonstrate that GSFL yields superior performance compared to state-of-the-art methods in both objective metrics and subjective visual quality. Despite this highly competitive performance, obtaining perfectly registered reference images in practical missions remains a common challenge. Therefore, our future work will extend this framework into an unsupervised or self-supervised training paradigm, further broadening its practicality and robustness in complex real-world fusion scenarios.

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